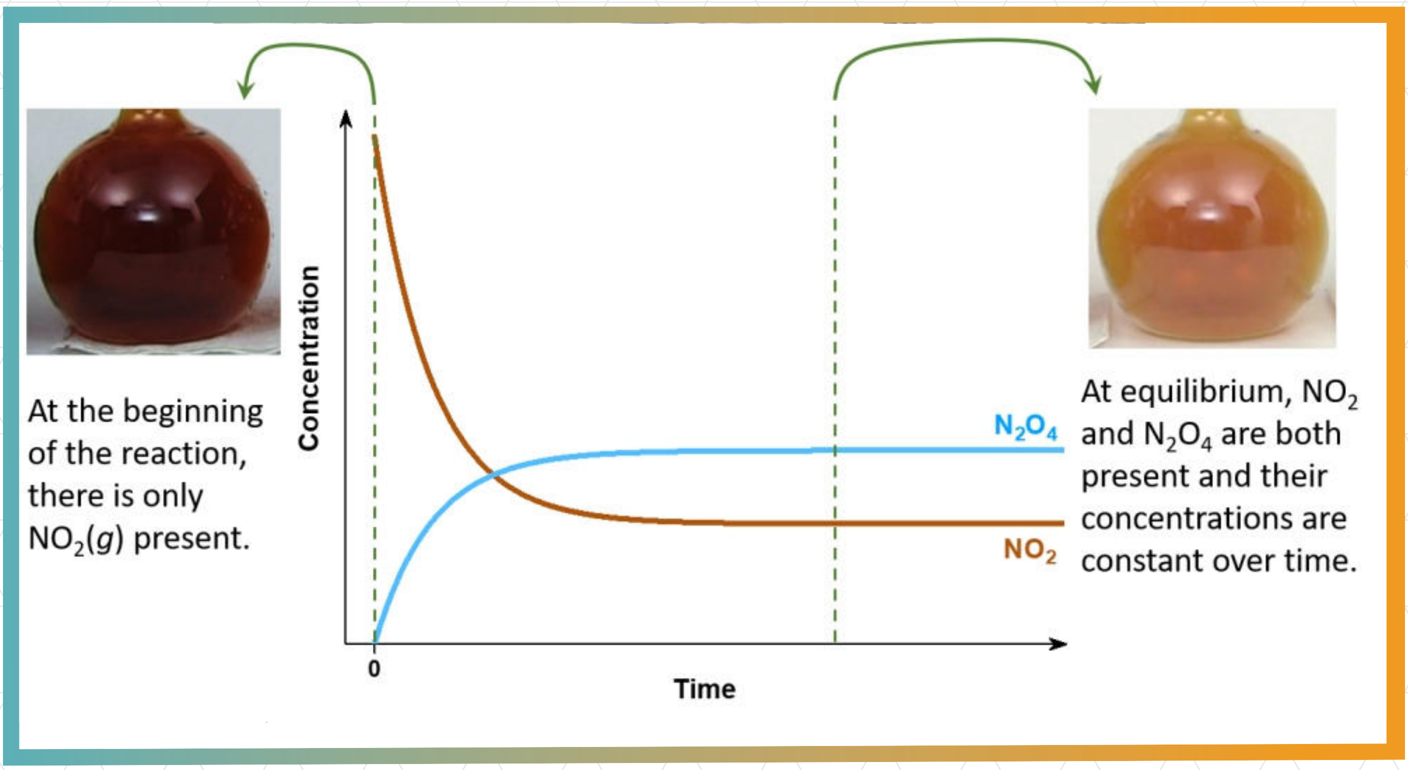
UNIT 01

CHEMICAL EQUILIBRIUM



CHEMICAL EQUILIBRIUM

Chemical equilibrium is a state in which the rates of forward and reverse reactions in a chemical system are equal, resulting in a constant concentration of reactants and products.

EXAMPLE:

1) The reaction between nitrogen gas (N_2) and hydrogen gas (H_2) to form ammonia (NH_3) exhibits chemical equilibrium:

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

2) The dissociation of hydrogen iodide (HI) into hydrogen (H_2) and iodine (I_2) is another example of chemical equilibrium:

$$2HI \rightleftharpoons H_2 + I_2$$

REVERSIBLE EQUILIBRIUM

Reversible equilibrium refers to a state in which a chemical reaction can proceed in both the forward and reverse directions. It occurs when the rates of the forward and reverse reactions are equal.

FORMULA

$$aA + bB \rightleftharpoons cC + dD$$

EXAMPLE:

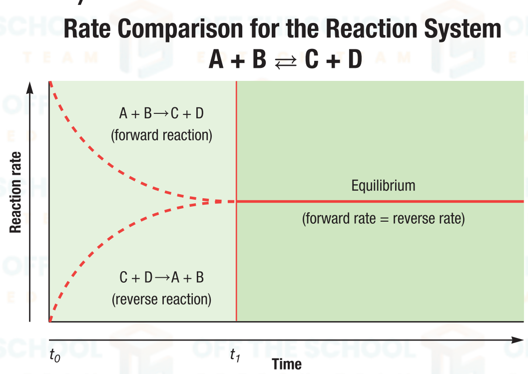
Consider the reaction between nitrogen gas (N_2) and hydrogen gas (H_2) to form ammonia (Nh_3):

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

DYNAMIC EQUILIBRIUM

Dynamic equilibrium describes a situation in which the **concentrations of reactants and products remain constant over time**, while the individual molecular-level processes continue to occur. It is characterized by a balance between forward and reverse reactions.

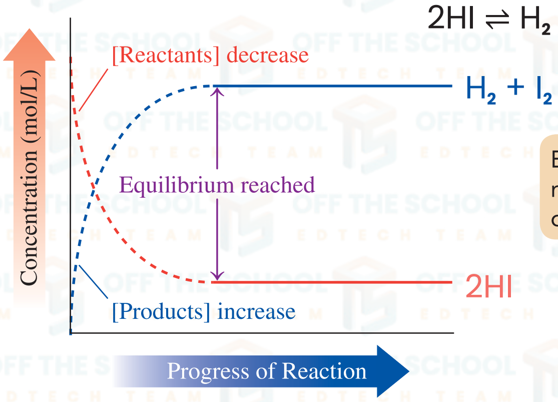
Equilibrium Reaction Systems Shown are reaction rates for the hypothetical equilibrium reaction system $A + B \rightleftharpoons C + D$. From the time A and B are mixed together at t_0, the rate of the forward reaction declines and the rate of the reverse reaction increases until both forward and reverse reaction rates are equal at t_1, when the equilibrium condition begins.



EXAMPLE:

Consider the dissociation of hydrogen iodide (HI) into hydrogen (H_2) and iodine (I_2) is another example of chemical equilibrium:

$$2HI \rightleftharpoons H_2 + I_2$$



Equilibrium is reached when there are no further changes in the concentrations of reactants and products.

MACROSCOPIC CHARACTERISTICS OF FORWARD AND REVERSE IN DYNAMIC EQUILIBRIUM

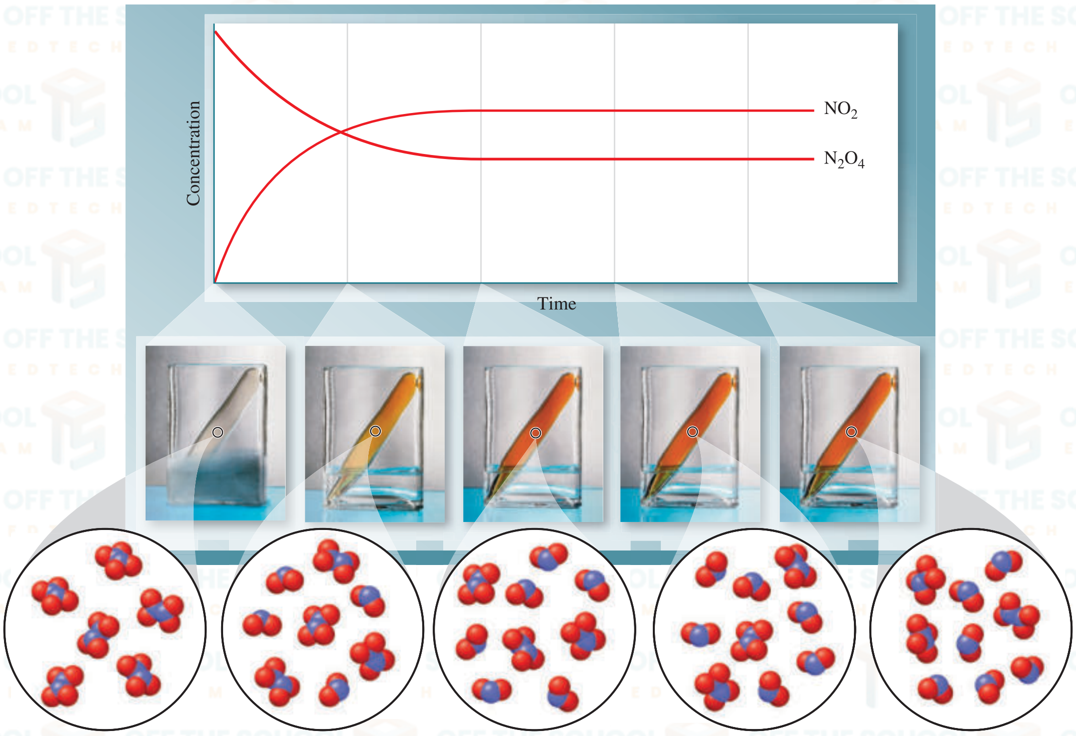
In a dynamic equilibrium, the forward and reverse processes exhibit the following macroscopic characteristics:

FORWARD REACTION

1. The forward process involves the conversion of reactants into products.
2. The concentration of reactants decreases as the forward process occurs.
3. The concentration of products increases as the forward process takes place.

REVERSE REACTION

1. The reverse process involves the conversion of products back into reactants.
2. The concentration of products decreases as the reverse process occurs.
3. The concentration of reactants increases as the reverse process takes place.



Reaction of colorless N_2O_4 to form brown NO_2. Initially, only N_2O_4 is present and only the forward reaction (decomposition of N_2O_4 to give NO_2) is occurring. As NO_2 forms, the reverse reaction (recombination of NO_2 to give N_2O_4) begins to occur. The brown color continues to intensify until the forward and reverse reactions are occurring at the same rate.

H_2, I_2, HI Equilibrium System

The violet color of the reactant iodine gas, I_2, gives an indication of the progress of the reaction between hydrogen and iodine. Both H_2 (reactant) and HI (product) are colorless gases. The darker the purple color in the flask, the less I_2 gas reacted, so the more reactants remain.

LAW OF MASS ACTION

INTRODUCTION

It was formulated independently by Cato Guldberg and Peter Waage in 1864.

STATEMENT

According to the law of mass action, **the rate of a chemical reaction is directly proportional to the product of the active masses of the reactants each term raised to its stoichiometric coefficients.**

$$aA + bB \rightleftharpoons cC + dD$$

Now, from the law of mass action, we know that the rate of forward reaction (Rf) and rate of backward reaction (Rb) will be:

$$R_f = k_f [A]^a[B]^b$$
$$R_b = k_b [C]^c[D]^d$$

Where kf and kb are the rate constants for the forward and backward reactions, respectively. After equilibrium is reached, we have

Previous Year's Questions

State the law of mass action, apply it on the following reversible reaction.
aA + bB ⇌ cC + dD
and derive the expression for equilibrium constant.

[2024]

$$R_f = R_b$$
$$K_f [A]^a[B]^b = K_b [C]^c[D]^d$$
$$\frac{Kf}{Kb} = \frac{[C]^c[D]^d}{[A]^a[B]^b} \text{ ---------- (i)}$$

Since the kf and kb are also constant at equilibrium, the ratio of the two is also a constant and is typically labelled as K or the equilibrium constant. Therefore, equation (i) is modified as:

$$\frac{Kf}{Kb} = K = \frac{[C]^c[D]^d}{[A]^a[B]^b} = \frac{[Product]}{[Reactant]}$$

All this leads to the modern definition of the "law of mass action" that the ratio of the multiplication of molar concentrations of products raised to the power of their stoichiometric coefficients to the multiplication of the molar concentrations of the reactants raised to the power of their stoichiometric coefficients is constant at constant temperature and is called as "equilibrium constant" This equation is also known as the "law of chemical equilibrium".

EQUILIBRIUM CONSTANT EXPRESSION

Equilibrium constant is a quantitative measure that expresses the ratio of product concentrations to reactant concentrations at a specific temperature, once a chemical reaction has reached equilibrium. Here are some general characteristics of the equilibrium constant expression:

Previous Year's Questions

Write three characteristics of equilibrium constant Kc?

[2023]

1) Kc only works in equilibrium.
2) Important characteristics of equilibrium constant expression are as follows Kc only works in equilibrium.
3) Kc is independent of reactant and product concentrations.
4) Kc is a balanced chemical equation coefficient. In a balanced chemical equation each

5) Kc represents equilibrium position. If Kc is larger than 1, the reaction is forward. If Kc is less than 1, the reaction is a reverse reaction.

6) Remember that equilibrium constant Kc is a ratio of reactant to product that is utilized to define chemical behavior.

IMPORTANCE OF EQUILIBRIUM CONSTANT

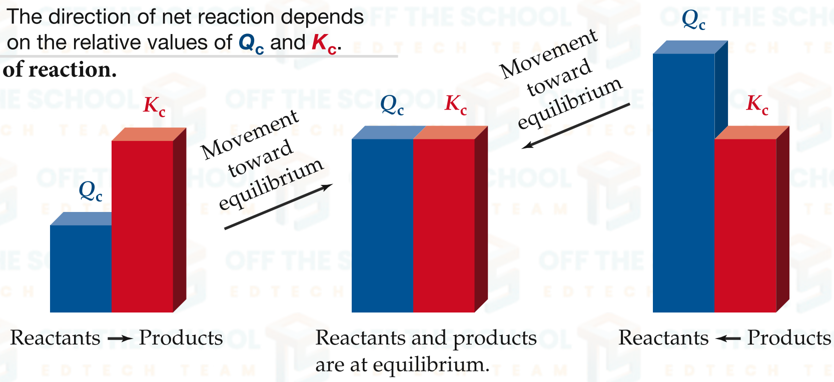
The value of Kc varies depending on the response. Kc isn't only a calculating constant. It affects both the direction and the extent of a chemical reaction.

1. DIRECTION OF A CHEMICAL REACTION

Determining the direction of reversible reactions is crucial for process optimization. The reaction quotient, Qc, helps predict the reaction's state at any given time using actual concentrations. By comparing Qc to Kc, informed decisions can be made for efficient production or adjustment of the reaction. Comparing Kc and Qc values predicts response direction. We have three categories:

The direction of net reaction depends on the relative values of Q_c and K_c.

Predicting the direction of reaction.



Movement toward equilibrium changes the value of Q_c until it equals K_c, but the **value of K_c** remains constant.

2. EXTENT OF CHEMICAL REACTION

At a certain temperature, the extent of a reaction measured. The magnitude of an equilibrium constant can predict the scope of a chemical reaction. As magnitude may be very high, very low, or moderate, so can be Extent of chemical reaction.

I. KC IS VERY SMALL:

Reactions with low Kc never finish. That is, maximum reactant concentration and minimum product concentration. These are called 'reverse or backward responses.

$F_{2(g)} \rightleftharpoons 2F_{(g)}$ $\quad$ $Kc = 7.4 \times 10^{-13}$ at 227°C

II. KC IS VERY LARGE:

Reactions with high values are virtually complete. That is, maximum product concentration and minimum reactant concentration. This type of reaction is known as 'Forward reaction'.

$2H_{2(g)} + O_{2(g)} \rightleftharpoons 2H_2O$ $\quad$ $Kc = 2.4 \times 10^{47}$ at 227°C

III. KC NEITHER VERY SMALL NOR VERY LARGE:

Reactions which have moderate value of Kc are considered to be at equilibrium. The concentration of reactants and products is almost same. For example:

$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ $\quad$ $Kc = 0.36$ at 25°C

1 ASSIGNMENT QUESTIONS

1 Write the equilibrium expression Kc of the following reactions:

1) $H_{2(g)} + I_{2(g)} \leftrightharpoons 2HI_{(g)}$
2) $N_{2(g)} + 3H_{2(g)} \leftrightharpoons 2NH_{3(g)}$
3) $2SO_{2(g)} + O_{2(g)} \leftrightharpoons 2SO_{3(g)}$
4) $CO_{(g)} + H_2O_{(g)} \leftrightharpoons CO_{2(g)} + H_{2(g)}$
5) $N_2O_{4(g)} \leftrightharpoons 2NO_{2(g)}$
6) $CH_3COOH_{(aq)} + H_2O_{(l)} \leftrightharpoons CH_3COO^-_{(aq)} + H_3O^+_{(aq)}$
7) $H_2CO_{3(aq)} \leftrightharpoons H_2O_{(l)} + CO_{2(g)}$
8) $2NaHCO_{3(s)} \leftrightharpoons Na_2CO_{3(s)} + H_2O_{(g)} + CO_{2(g)}$
9) $CaCO_{3(s)} \leftrightharpoons CaO_{(s)} + CO_{2(g)}$
10) $Fe_3O_{4(s)} + 4H_{2(g)} \leftrightharpoons 3Fe_{(s)} + 4H_2O_{(g)}$
11) $CH_3OH_{(g)} + H_{2(g)} \leftrightharpoons CH_{4(g)} + H_2O_{(g)}$
12) $C_2H_{4(g)} + H_{2(g)} \leftrightharpoons C_2H_{6(g)}$
13) $N_2O_{(g)} + NO_{(g)} \leftrightharpoons N_{2(g)} + 2O_{2(g)}$
14) $2NO_{(g)} + Cl_{2(g)} \leftrightharpoons 2NOCl_{(g)}$
15) $SO_{2(g)} + Cl_{2(g)} \leftrightharpoons SO_2Cl_{2(g)}$
16) $H_2S_{(g)} + Cl_{2(g)} \leftrightharpoons 2HCl_{(g)} + S_{(g)}$
17) $PCl_{5(g)} \leftrightharpoons PCl_{3(g)} + Cl_{2(g)}$
18) $CO_{2(g)} + H_{2(g)} \leftrightharpoons CH_3OH_{(g)}$
19) $2CO_{(g)} + O_{2(g)} \leftrightharpoons 2CO_{2(g)}$
20) $2H_{2(g)} + O_{2(g)} \leftrightharpoons 2H_2O_{(g)}$

$$\frac{Kf}{Kb} = aA + bB \rightleftharpoons cC + dD \qquad K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b} = \frac{[Product]}{[Reactant]}$$

2 Balance each reaction and write its Kc expression:

(a) $NO(g) + O_2(g) \rightleftharpoons N_2O_3(g)$
(b) $SF_6(g) + SO_3(g) \rightleftharpoons SO_2F_2(g)$
(c) $SClF_5(g) + H_2(g) \rightleftharpoons S_2F_{10}(g) + HCl(g)$

3 Balance each reaction and write its Kc expression:

(a) $C_2H_6(g) + O_2(g) \rightleftharpoons CO_2(g) + H_2O(g)$
(b) $CH_4(g) + F_2(g) \rightleftharpoons CF_4(g) + HF(g)$
(c) $SO_3(g) \rightleftharpoons SO_2(g) + O_2(g)$

3 Balance each reaction and write its Kc expression:

(a) $\frac{1}{2}S_2(g) + H_2(g) \rightleftharpoons H_2S(g)$
(b) $5H_2S(g) \rightleftharpoons 5H_2(g) + \frac{5}{2}S_2(g)$
$2H_2S(g) \rightleftharpoons 2H_2(g) + S_2(g)$

4 Balance each reaction and write its Kc expression:

(a) $NO(g) + H_2(g) \rightleftharpoons \frac{1}{2}N_2(g) + H_2O(g)$
(b) $2N_2(g) + 4H_2O(g) \rightleftharpoons 4NO(g) + 4H_2(g)$
(c) $2NO(g) + 2H_2(g) \rightleftharpoons N_2(g) + 2H_2O(g)$

5 Balance each reaction and write its Kc expression:

(a) $Na_2O_2(s) + CO_2(g) \rightleftharpoons Na_2CO_3(s) + O_2(g)$
(b) $H_2O(l) \rightleftharpoons H_2O(g)$
(c) $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$

6 Balance each reaction and write its Kc expression:

(a) $H_2O(l) + SO_3(g) \rightleftharpoons H_2SO_4(aq)$
(b) $KNO_3(s) \rightleftharpoons KNO_2(s) + O_2(g)$
(c) $S_8(s) + F_2(g) \rightleftharpoons SF_6(g)$

7 Determine the Kc unit of the following reactions:

(a) $2KClO_3(s) \rightleftharpoons 2KCl(s) + 3O_2(g)$
(b) $2PbO(s) + O_2(g) \rightleftharpoons 2PbO_2(s)$
(c) $I_2(s) + 3XeF_2(s) \rightleftharpoons 2IF_3(s) + 3Xe(g)$

8 Determine the Kc unit of the following reactions:

(a) $MgCO_3(s) \rightleftharpoons MgO(s) + CO_2(g)$
(b) $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(l)$
(c) $HNO_3(l) + ClF(g) \rightleftharpoons ClONO_2(g) + HF(g)$

CASE # 01 : (Kc expression and active mass)

1) Consider the following equilibrium system: $2NO + Cl_2 \rightleftharpoons 2NOCl$. At a certain temperature, the equilibrium concentrations are as follows: $[NO] = 0.184$ M, $[Cl_2]=0.165$ M, $[NOCl]=0.060$ M. What is the equilibrium constant for this reaction?
For the reaction: $MgCl_2 + \frac{1}{2} O_2 \rightleftharpoons 2MgO + Cl_2$. The equilibrium constant was found to be 3.86 at a certain temperature. If $[O_2] = 0.560$ M at equilibrium, what is the concentration of $Cl_2(g)$?
3) Consider the equilibrium: $CO_{(g)} + H_2O_{(g)} \rightleftharpoons H_{2(g)} + CO_{2(g)}$.
   a) Write an equilibrium expression for this reaction.
   b) If $[CO] = 0.200M$, $[H_2O] = 0.500M$, $[H_2] = 0.32M$ and $[CO_2] = 0.42M$, find K_c.
4) Hydrogen sulfide decomposes according to the equation: $2H_2S_{(g)} \rightleftharpoons 2H_{2(g)} + S_{2(g)}$.
   a) Write an equilibrium expression for this reaction.
   b) At equilibrium, the concentrations of each gas are as follows: $[H_2S] = 7.06\times10^{-3}$ M, $[H_2] = 2.22\times10^{-3}$ M and $[S_2] = 1.11\times10^{-3}$M. What is K_{eq}?
5) Given the following system in equilibrium: $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO^{3(g)}$
   a) Write an equilibrium expression for the reaction.
   b) If Kc= 85.0, would you expect to find more reactants or products at equilibrium? Why?
   c) If $[SO_2] = 0.0500$ M and $[O_2] = 0.0500$M, what is the concentration of SO_3 at equilibrium?
6) For the reaction, $SO_{2(g)} + NO_{2(g)} \rightleftharpoons SO_{3(g)} + NO_{(g)}$ determine the value of K_c when the equilibrium concentrations are: $[SO_2] = 1.20$ M, $[NO_2] =0.60$ M, $[NO]=1.6$ M, and $[SO_3] = 2.2$ M.
7) Consider the following reaction: $CO_{(g)} + H_2O_{(g)} \rightleftharpoons H_{2(g)} + CO_{2(g)}$; K=1.34 If the $[H_2O] =0.100$ M, $[H_2] = 0.100$ M, and $[CO_2] = 0.100$ M at equilibrium, what is the equilibrium concentration of CO?
8) For the reaction, $CO + 3H_2 \rightleftharpoons CH_4 + H_2O$, Calculate Kc from the following equilibrium concentrations: $[CO] = 0.0613$ M; $[H_2] = 0.1839$ M; $[CH_4] = 0.0387$ M; $[H_2O] = 0.0387$ M.
9) A quantity of HI was sealed in a tube, heated to 425°C and held at this temperature until equilibrium was reached. The concentration of HI in the tube at equilibrium was found to be 0.0706 mol/L. Calculate the equilibrium concentration of H_2 (and I_2). For the gas-phase reaction. The Kc for the reaction is 54.6.
$$H_2 + I_2 \rightleftharpoons 2HI$$
10) If the equilibrium concentrations found in the reaction $A_{(g)} + 2B_{(g)} \rightleftharpoons 2C_{(g)}$ are $[A] = 0.025$ M, $[B] = 0.15$ M, and $[C] = 0.55$ M, calculate the value of K_c.
11) PCl_5, PCl_3 and Cl_2 are at equilibrium at 500K in a closed container and their concentrations are 0.8×10^{-3} mol.L^{-1}, 1.2×10^{-3} mol.L^{-1} and 1.2×10^{-3} mol.L^{-1} respectively. The value of Kc for the reaction: $PCl_5\ (g) \Leftrightarrow PCl_{3\ (g)} + Cl_2\ (g)$ will be?
12) $2NOBr_{(g)} \rightleftharpoons 2NO_{(g)} + Br_{2(g)}$: at 727°C, the equilibrium concentration of NO is 1.29 M, Br_2 is 10.52 M, and NOBr is 0.423 M.

CASE #02: (Predict the direction Qc)

1) The value of Kc for the reaction is 1×10^{-4}
$$H_2 + I_2 \rightleftharpoons 2HI$$
At a given temperature, the molar concentration of reaction mixture is HI = 2×10^{-5} mol.dm^{-3}, $H_2 = 1\times10^{-5}$ mol dm^{-3} and 1×10^{-5} mol dm^{-3}. Predict the direction of the reaction.
2) What is the Qc value for this equation? Which direction will the reaction shift if Kc = 1.0?
$$CO_{(g)}+H_2O_{(g)} \rightleftharpoons CO_{2\ (g)} + H_{2(g)}$$
- $[CO_2] = 2.0$ M
- $[H_2] = 2.0$ M
- $[CO] = 1.0$ M
- $[H_2O] = 1.0$ M